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G. 自编码器模型细节

我们以对抗方式训练所有自编码器模型，遵循 [23]，使得基于块的判别器 D_ψ 被优化以区分原始图像与重建图像 $\mathcal{D}(\mathcal{E}(x))$ 。为避免任意缩放的潜空间，我们正则化潜变量 z 使其零中心化，并通过引入正则化损失项 L_{reg}. 获得较小方差。我们研究两种不同的正则化方法：(i) 在 $q_{\mathcal{E}}(z|x) = \mathcal{N}(z;\mathcal{E}_\mu,\mathcal{E}_{\sigma^2})$ 与标准正态分布 $\mathcal{N}(z;0,1)$ 之间使用低权重 Kullback-Leibler 项，如标准变分自编码器 [46,69] 中那样；(ii) 通过向量量化层正则化潜空间，学习包含 $|\mathcal{Z}|$ 个不同示例样本的码本 [96]。为获得高保真重建，我们对两种情形仅使用非常小的正则化，即要么以因子 $\sim 10^{-6}$ 加权 $\mathbb{KL}$ 项，要么选择高码本维度 $|\mathcal{Z}|$. 。训练自编码模型 $(\mathcal{E},\mathcal{D})$ 的完整目标函数为：

$$L_{\text{Autoencoder}} = \min_{\mathcal{E},\mathcal{D}} \max_{\psi} \Big(L_{rec}(x,\mathcal{D}(\mathcal{E}(x))) - L_{adv}(\mathcal{D}(\mathcal{E}(x))) + \log D_\psi(x) + L_{reg}(x;\mathcal{E},\mathcal{D}) \Big) \tag{25}$$

潜空间中的扩散模型训练 注意，在学习到的潜空间上训练扩散模型时，我们再次区分学习 $p(z)$ 或 $p(z|y)$ 的两种情况（第 4.3 节）：(i) 对于 KL 正则化的潜空间，我们从 $z = \mathcal{E}_\mu(x)+\mathcal{E}_\sigma(x)\cdot\varepsilon =: \mathcal{E}(x)$, 采样，其中 $\varepsilon \sim \mathcal{N}(0,1)$ 。当重新缩放潜变量时，我们估计逐分量方差

$$\hat{\sigma}^2 = \frac{1}{bchw}\sum_{b,c,h,w}(z^{b,c,h,w} - \hat{\mu})^2$$

从数据中的第一批次估计，其中 $\hat{\mu} = \frac{1}{bchw}\sum_{b,c,h,w} z^{b,c,h,w}$ 。$\mathcal{E}$ 的输出被缩放，使得重新缩放后的潜变量具有单位标准差， *i.e.* $z \leftarrow \frac{z}{\hat{\sigma}} = \frac{\mathcal{E}(x)}{\hat{\sigma}}$ 。(ii) 对于 VQ 正则化的潜空间，我们在量化层之前提取 z ，并将量化操作吸收到解码器中，即它可以被解释为 $\mathcal{D}$. 的第一层

H. 附加定性结果

最后，我们为我们的风景模型（图 12、23、24 和 25）、类条件 ImageNet 模型（图 26-27）以及 CelebA-HQ、FFHQ 和 LSUN 数据集上的无条件模型（图 28-31）提供附加定性结果。与第 4.5 节中的修复模型类似，我们还在 512^2 图像上直接微调了第 4.3.2 节的语义风景模型，并在图 12 和图 23 中展示了定性结果。对于在相对较小数据集上训练的模型，我们还在图 32-34 中展示了来自我们模型的样本在 VGG[79] 特征空间中的最近邻。